

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

BIRD CAN FLY OR NOT

Aim:

To write a Prolog program to determine whether a particular bird can fly.

Objective:

To demonstrate logical inference using facts and rules.

Algorithm:

- ☐ **Start.**
- ☐ **Define birds.**
- ☐ **Define birds that cannot fly.**
- ☐ **Define the rule that a bird can fly if it is a bird and is not flightless.**
- ☐ **Query the required bird.**
- ☐ **Display whether it can fly.**
- ☐ **Stop.**

Flowchart:

START

|

v

Enter Bird

|

v

Is it a Bird?

/ \

No Yes

| |

Cannot Is it Flightless?

**Fly / **

Yes No

| |

Cannot Can

Fly Fly

\ /

STOP

Prolog Program:

% Bird Flying System

bird(eagle).

bird(parrot).

bird(pigeon).

bird(crow).

bird(ostrich).

bird(penguin).

cannot_fly(ostrich).

cannot_fly(penguin).

can_fly(Bird)

:bird(Bird),

\+ cannot_fly(Bird).

Output:

```
78 ?- consult(['c:/Users/Niharika Preejesh/OneDrive/Documents/bird can fly or not.pl']).
true.

79 ?- check_flying(parrot).
parrot can fly.
true
```

Result:

Thus, the Prolog program successfully determines whether a given bird can fly.